

HDOC DAY -12

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BATCH-15

Problem Statement:

Write a menu-driven C program using switch-case to input a positive integer and perform the following operations based on the user's choice:

1. Find the sum of squares of its digits using a while loop.
2. Find the sum of cubes of its digits using a for loop.
3. Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop

Aim

To write a menu-driven C program using **switch-case** to perform operations on the digits of a positive integer using **while, for, and do-while loops**

Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu:
 - 1. Sum of squares of digits
 - 2. Sum of cubes of digits
 - 3. Difference between sum of cubes and sum of squares
4. Input the user's choice.
5. Use switch-case to perform the selected operation.
6. Display the result.
7. Stop.

No. Input Choice Output

1	123	1	14
2	123	2	36
3	123	3	22
4	123	4	Invalid choice

>C CODE:

```
#include <stdio.h>

int main(){
    int n,t,d,ch,s=0,c=0;
    printf("Enter number: ");
    scanf("%d",&n);
    printf("1.Square 2.Cube 3.Difference\nEnter choice: ");
    scanf("%d",&ch);
    t=n;
    switch(ch){
    case 1:
        while(t){d=t%10;s+=d*d;t/=10;}
        printf("Sum = %d",s); break;
    case 2:
        for(;t/=10){d=t%10;c+=d*d*d;}
        printf("Sum = %d",c); break;
    case 3:
        do{d=t%10;s+=d*d;c+=d*d*d;t/=10;}while(t);
        printf("Difference = %d",c-s); break;
    default: printf("Invalid choice");
    }
    return 0;
```

